

Prakalya Sampath

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SUMMARY

- **Full-Stack & AI Developer:** Skilled in Java Spring Boot and React.js, with hands-on experience integrating Generative AI (Gemini API / Spring AI), Machine Learning models, and Deep Learning frameworks into high-performance, scalable applications.
- **AWS-focused Solutions Architect:** B.Tech AI & Data Science student (8.6 CGPA) experienced in building event-driven, serverless solutions using AWS Lambda, S3, and DynamoDB, alongside deploying intelligent ML/DL pipelines in the cloud.
- **Data Science & Core ML Intern:** Former Machine Learning Intern with proven experience engineering predictive models, optimizing neural networks, and refining data pipelines for measurable performance gains and data-driven insights.
- **Cloud & DevOps Enthusiast:** Experienced with Amazon EKS, containerized deployments (Docker), and automated CI/CD pipelines using GitHub Actions to streamline both software and MLOps workflows.
- **Relentless Problem Solver:** Solved 500+ DSA problems across LeetCode and GeeksforGeeks, maintaining a 100-day consistency streak on LeetCode with strong algorithmic thinking.

TECHNICAL PROFICIENCIES

Programming Languages: Java, Python, Cpp, SQL

Data Structures & Algorithms: Arrays, Strings, Linked List, Queue, Stacks, Hashing, Trees

Backend Technologies: Spring Boot, Spring Security, Hibernate, RESTful API Design, JWT Authentication

Frontend Development: React.js, Bootstrap (Basics)

Databases: MySQL, PostgreSQL

Cloud & DevOps: AWS (IAM,EC2, S3, Lambda, Dynamo DB,SNS,Amazon EKS), Github Actions(CI&CD), Git/GitHub.

ML,DL & Libraries: Pandas, NumPy, Scikit-learn,Matplotlib

Tools & Platforms: Git, GitHub, IntelliJ IDEA, Visual Studio Code, Postman,Google Colab

PROJECTS

1.Serverless AI Image Processing & Alerting System-AWS (S3, Lambda, Rekognition, DynamoDB, SNS)

- Achieved a 100% reduction in operational overhead by migrating from dedicated servers to an AWS Lambda-based event-driven architecture, eliminating manual server patching and maintenance.
- Optimized cloud expenditures to near-zero idle costs by designing an architecture where AWS services communicate via events, triggering execution only upon image uploads to Amazon S3.
- Integrated Amazon Rekognition to automate real-time image analysis with 90%+ object detection confidence, enabling instant metadata extraction without human intervention.
- Engineered a high-performance Metadata Store in DynamoDB, ensuring sub-second retrieval of analysis history and image tags for seamless user access.
- Automated real-time security alerts using Amazon SNS, triggering instant email notifications for specific detected objects to enhance system responsiveness.

2.Cloud Infrastructure FinOps & MLOps Ingestion Pipeline-Python, LightGBM, FastAPI, Kafka, Redis, Docker, Kubernetes, MLflow

- Built a real-time cloud infrastructure cost forecasting platform using LightGBM, Kafka, Redis, and FastAPI to predict Kubernetes-based infrastructure spending from telemetry streams.
- Designed scalable streaming pipelines using Apache Kafka for continuous telemetry ingestion and integrated Redis caching for low-latency inference.
- Engineered advanced time-series and utilization-based features including rolling averages, lag metrics, and workload efficiency ratios to improve regression performance
- Containerized ML microservices with Docker and implemented monitoring pipelines using Prometheus and Grafana for production observability and drift tracking.

3. Emergency Service Booking Platform - React, Spring Boot, MySQL, AWS(EC2,S3,RDS,EKS),Github actions(CI&CD),Docker

- Reduced database query latency by 85% through rigorous MySQL schema normalization and optimized indexing, ensuring high-speed data retrieval for concurrent booking requests.
- Architected a high-reliability booking system for non-emergency medical transport, implementing real-time availability tracking to handle peak traffic spikes effectively.
- Developed secure, stateless authentication using JWT and Spring Security, protecting sensitive patient data while managing high volumes of concurrent user sessions.
- Streamlined deployment cycles by implementing a CI/CD pipeline with GitHub Actions, automating the build and test process for containerized services deployed to Amazon EKS.
- Ensured 99.9% application availability by leveraging AWS RDS for scalable database management and Amazon S3 for reliable frontend hosting.

WORK EXPERIENCE

Machine Learning Intern | Cognifyz Technologies

Engineered a Restaurant Recommendation System using content-based filtering and proximity calculations to deliver personalized suggestions

- Built and fine-tuned supervised ML models to predict user ratings with high accuracy through effective feature engineering.
- Designed scalable data pipelines to preprocess 50,000+ reviews, handling missing values and outliers for clean model inputs.
- Managed the full ML lifecycle, from Exploratory Data Analysis (EDA) to model evaluation using RMSE and MAE metrics.
- Improved recommendation reliability by reducing model errors through hyperparameter tuning.

CERTIFICATIONS

IBM Java Developer Certification , Microsoft AI and ML Engineering , AWS Skill Builder –Cloud Practitioner Essentials (Hands-on Training),Postman API Fundamentals Student Expert , NPTEL Design Thinking -A primer (Elite)

EDUCATIONAL QUALIFICATIONS

B.Tech–Artificial Intelligence & Data Science | Sri Krishna College of Engineering & Technology | CGPA:8.6 | 2024-Present

Higher Secondary Education (Class XII) | Vellalar Matric Higher Secondary School, Erode | 96% | 2023-2024

Secondary Education (Class X) |Vellalar Matric Higher Secondary School, Erode | 95% | 2022-2023